Mathematics for Computing

Assignment 1 : Set Theory

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Assignment No.1

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Introduction

- **What is a set?**

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  - Example of set:{a,e,i,o,u }
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  - Example of set:{a,e,i,o,u }
  - Example of a set: Empty set (no items): { } or ϕ
  - Example of not a set: Set of all tall people in this room
  - Example of not a set:The collection of the best football players in the world.

Set Builder Form

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   $$Q_1 = \{(x, y) \mid x, y \in \mathbb{R} \; and \; x > 0 \; and \; y > 0 \}$$

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$$S = \{x \mid x^2 < 50 \ \forall\ x \in \mathbb{Z}\}$$

Sets used in AI and ML Domain